

SUMMARY

Senior Backend Engineer with 6 years of experience architecting and building highly available, scalable cloud systems processing millions of daily transactions. A collaborative builder and critical thinker who values simplicity and low-risk solutions. Led architecture design for 12+ technical designs across microservices, Machine Learning systems, and real-time pipelines. Combines an innovative, highly adaptive mindset with deep engineering accountability. Active technical blogger at cuongkane.com.

EXPERIENCE

- **Parcel Perform – Delivery Experience SaaS Platform** Ho Chi Minh City, Vietnam
Senior Backend Engineer Aug 2024 - Present
 - **Product Scope:** Serving enterprise e-commerce brands across Europe and North America. Performance Alerting - proactive email notifications on delivery metrics; Reporting & Analysis - carrier benchmarking dashboards and trend analysis; AI Recommendation - data-driven carrier selection and delivery optimization insights; Performance Aspiration - SLA framework measuring actual vs expected transit time across warehouse, linehaul, and last-mile phases.
 - **System Architecture & Design:** Authored 8+ technical designs spanning APIs, event-driven pipelines, batch orchestration, async processing, and authentication systems across 15 microservices; reviewed 5+ peer designs.
 - **Performance Aspiration Platform:** Designed and built 2 new microservices and a stateful Flink pipeline processing around 1K events/s for automated SLA compliance measurement. Tackled challenges around stateful stream processing with broadcast-state rule management, high-throughput evaluation at scale, and reliable dual-sink data syncing to PostgreSQL and ClickHouse for API queries and BI dashboards.
 - **Timezone-Aware Analytics Pipeline:** Redesigned the AI Recommendations pipeline to deliver timezone-aware insights, ensuring all analytics and AI-generated reports reflect each customer's local business day instead of UTC.
 - **Microservices & Database Splitting:** Proposed and engineered decomposition of 2 monoliths into microservices with zero-downtime database splitting, enhancing modularity and enabling autonomous team deployments, improving team velocity by 30%.
 - **Large-Scale Database Migrations:** Executed zero-downtime PostgreSQL migrations on tables with billions of records using batch processing strategies, ensuring continuous service availability during schema evolution.
 - **High-Availability Cloud Systems:** Owned and scaled mission-critical analytics platform (ClickHouse/Dagster); accelerated dashboard loading by 30% through static file caching.
 - **AI/ML Solutions:** Pioneered building AI agents using LangChain & LangGraph following ReAct architecture with AWS Bedrock; reduced LLM costs by 20% via strategic model selection while automating Root Cause Analysis workflows.
 - **Developer Productivity Tooling:** Engineered 5+ Claude Code skills adopted company-wide (15+ engineers), improving review speeds by 20%; resolved context window bloating and automated doc maintenance.
 - **AI-Assisted Software Development:** Initiated Specification-Driven Development (SDD) workflow within the team, inspiring adoption across other teams; maintained top productivity in the team across both AI and non-AI areas; and transformed traditional microservices into AI-ready code repositories with structured documentation and conventions.
 - **Release & Deployment Strategy:** Orchestrated canary deployment protocols and A/B testing pipelines, achieving zero-downtime releases and minimizing production risk through incremental delivery.
 - **Observability & Sustainable Growth:** Improved system reliability by defining strict SLA/SLO targets; leveraged p99 latencies and Kafka lag metrics to automate capacity planning and preemptively resolve bottlenecks.
 - **Mentorship & Knowledge Sharing:** Accelerated team growth by onboarding 5+ engineers and mentoring a developer to promotion-readiness; fostered engineering excellence by delivering 15+ company-wide technical sessions on System Design, AI/LangChain, and best practices.
 - **Engineering Leadership:** Strengthened engineering standards by promoting engineering culture, showcasing clean code principles and software best practices, and conducting 8+ technical interviews.
- **Parcel Perform – Delivery Experience SaaS Platform** Ho Chi Minh City, Vietnam
Backend Engineer Jul 2022 - Aug 2024

- **Product Scope:** Carrier Integration - parcel event scraping, event mapping, and scheduled scraping for 1,000+ carriers; Scheduler - periodic parcel scraping management; Booking & Returns - end-to-end return label generation, multi-carrier booking, and self-service returns; Collection Point - PUDO management for click-and-collect and locker workflows.
- **Carrier Integration & Event Pipeline:** Extended the carrier integration platform supporting 1,000+ carriers globally — adding new scrapers for parcel event ingestion, extending event mapping to normalize carrier statuses into a unified lifecycle taxonomy, and implementing scheduled scraping orchestration for rate-limited event collection.
- **Engineering Excellence:** Delivered features weekly with zero critical bugs by rigorously applying Test-Driven Development (TDD) and effective task prioritization.
- **Distributed Systems & Orchestration:** Architected a logic-intensive Public API for Booking & Returns, orchestrating complex multi-carrier workflows.
- **Sustainable Pipeline Architecture:** Accelerated data processing by parallelizing 8K+ postal code tasks using optimized Celery canvas patterns (chain/chord) and worker configurations; safeguarded external dependencies with Celery-based rate limiting.
- **Extensible Integration Architecture:** Designed extensible integration framework supporting 50+ carriers; validated requirements through usage testing to ensure scalability for future logistics partners.
- **Real-Time Data Pipelines:** Designed ML-based carrier identification pipeline processing millions of daily events across 7 microservices, approved by CTO.
- **High-Volume Data Pipelines:** Scaled data pipeline to process 5M+ daily events for mission-critical tracking using Apache Flink and optimized microservices.
- **Email Extraction Platform:** Engineered high-throughput email extraction engine using Kafka CDC; system has processed shipment notifications reliably in production for 3+ years.
- **Sync-to-Async Reliability Pattern:** Ensured booking reliability by architecting a sync-to-async fallback pattern across 5 microservices using Kafka to handle component failures.
- **System Reliability & Incident Response:** Restored 1.5M records during critical incidents within 24 hours by developing custom recovery scripts and resolving Redis cluster hotspots under 10x traffic load.
- **Large-Scale Data Migrations:** Secured data consistency during large-scale migrations by implementing incremental update patterns, optimizing SQL indexing, and managing autovacuum constraints.
- **Testing & Developer Experience:** Defined and spread company-wide testing standards (AAA pattern) and drove adoption of modern tooling (Poetry, Ruff).

• Cinnamon AI Labs

Ho Chi Minh City, Vietnam

Software Engineer

Aug 2020 - Jul 2022

- **Product Scope:** AI outsourcing company specializing in Computer Vision and NLP. Encapsulated and shipped AI products to Japanese enterprise clients (Deloitte, Bridgestone, Daikin, Sumitomo) based on customer requirements and supplied data.
- **Enterprise Client Delivery:** Deployed AI-powered document processing systems for Japanese enterprise clients, ensuring strict SLA compliance and driving client success in AI automation.
- **Multi-Framework Backend Development:** Delivered 30+ production APIs using Flask, FastAPI, and Django, strictly adhering to SOLID principles and Repository patterns.
- **Multi-Cloud Infrastructure:** Managed multi-cloud infrastructure (AWS, Azure, GCP) and containerized applications with Docker to ensure consistent cross-cloud deployments.
- **ML-to-Production Pipeline:** Bridged gap between research and production by converting experimental ML models into scalable, SLA-compliant API services.
- **Asynchronous Model Execution:** Implemented Celery-based background processing to execute ML models asynchronously and persist results to the database, ensuring high system responsiveness.

PROJECTS

- **Codlis - AI Code Visualization (Master's Thesis, H2 2025) (codlis.com):** Built multi-agent LLM system for code comprehension using LangChain/LangGraph and Knowledge Graphs. User study showed 2.9x higher task success rate and 29% cognitive load reduction compared to traditional methods.
- **SweatCharge - Sports Club Management (H1 2025) (sweatcharge.com):** Developed full-stack sports management platform with role-based SSO and automated billing; currently servicing active production users. Implemented fuzzy matching algorithms for player association.
- **Hasiti - AI Hairstyle Simulator (Bachelor's Thesis, 2021):** Built real-time hairstyle simulation app using fine-tuned CNNs for style transfer. Architected full-stack solution connecting React Native mobile client with Python/Flask AI inference backend.

CERTIFICATIONS

- **AWS Certified Solutions Architect - Associate:** Amazon Web Services (Valid: Jan 2025 – Jan 2028)

EDUCATION

- **Master of Science in Computer Science** Sept. 2021 – Sept. 2025
HCM University of Science *Ho Chi Minh City, Vietnam*
 - **Research Focus:** AI/ML, Big Data, and Human-Computer Interaction under post-doctoral supervision. Applied analytical experimentation methods for metric-driven evaluation.
- **Honor Program - Bachelor of Science in Information Technology** Sept. 2017 – Sept. 2021
HCM University of Science *Ho Chi Minh City, Vietnam*
 - **Honor Program:** Top 70 students selected from 1,000+ candidates. Comprehensive curriculum spanning web/mobile development, computer vision, image processing, software engineering, and scientific research methods.
 - **Technical Foundation:** Developed fundamentals in algorithms, data structures, OOP principles, and hardware/software architecture through rigorous coursework.

AWARDS & RECOGNITION

- **Parcel Perform (2022-Present):** Member of the Month (3x), Best Presentation Award, AVA AI Award for AI adoption, 2nd Prize – AI Adoption Champion (Individual), Freddy Fast Award.
- **Cinnamon AI Labs (2020-2022):** Employee Recognition Award (Monthly) for outstanding technical contributions.
- **Faculty of IT Academic Excellence (2018):** Top 10 students with highest GPA (9.08/10) among 1,000+ students.

TECHNICAL SKILLS

- **Languages:** Python (expert), TypeScript/JavaScript, Scala, Java, SQL
- **Frameworks:** Django (expert), FastAPI, LangChain/LangGraph, Next.js/React
- **Libraries:** Pydantic (expert), Django-Ninja, DRF
- **Distributed Systems:** Microservices, Event-Driven (Kafka, RabbitMQ), High-Availability, Outbox Pattern, Idempotency, Stateful Job
- **Data Pipelines:** Apache Flink, Dagster, Celery (expert)
- **Databases:** PostgreSQL, Redis, ClickHouse
- **Infrastructure:** AWS, Docker, CI/CD, Observability (Grafana, Sentry, Loki)
- **Software Development Concepts:** SDD, TDD, Agile, Technical Planning, Legacy Code Refactoring (expert)
- **Development Tools:** NeoVim (expert), Tmux, Claude Code, Claude Squad, Git (expert)

LANGUAGES

- **English:** B2 Upper Intermediate – Aptis ESOL (British Council) 176/200 **Vietnamese:** Native